



# UNIVERSITI KEBANGSAAN MALAYSIA

*The National University of Malaysia*

**COURSE:**

TTTC2343

SISTEM ROBOT CERDAS

**SYSTEM:**

Autonomous Warehouse Inventory and Item Delivery Robot

**GROUP:**

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## 1. Basic Industrial Problem

Warehouse management systems today face increasing complexity due to the rapid growth of e-commerce and the demand for faster order fulfillment. Traditional warehouse and production environments still rely heavily on manual internal logistics, where human workers are responsible for transporting goods between multiple operational zones. This approach leads to slow material flow, high dependency on human labour, and reduced scalability when order volumes increase.

In addition, warehouse operations involve multiple repetitive and physically demanding processes such as item picking, packing, sorting, inventory tracking, and goods distribution. These human-dependent workflows require high coordination and precision, but they are vulnerable to fatigue-related performance decline, which reduces consistency and increases the likelihood of errors in order fulfillment and inventory management. Furthermore, conventional manual systems struggle to handle real-time inventory updates, dynamic order changes, and unexpected disruptions such as congestion and obstacles, resulting in workflow bottlenecks and reduced operational efficiency.

Recent industrial developments show the integration of automation technologies such as Robotic Process Automation (RPA) and autonomous mobile robotic systems in warehouse environments. For example, industry applications have demonstrated that automated systems can improve order processing efficiency, reduce human workload, and enhance operational accuracy in logistics operations (Oh et al. 2024). However, despite these advancements, many warehouse facilities still rely partially or fully on manual processes, creating a gap between modern logistics requirements and existing operational capabilities.

Therefore, the main industrial problem is the inefficiency, inconsistency, and limited scalability of manual warehouse management processes in handling high-volume, real-time, and accuracy-critical operations.

### Reference :

*Oh, W. X., Selamat, H. & Muslim, M. T. 2024. Autonomous mobile robot for transporting goods in warehouse and production. Advances in Intelligent Manufacturing and Robotics, Lecture Notes in Networks and Systems 845: 555-565.*

*Tayab, A. & Li, Y., 2024. The Contribution of the RPA Technology in Enhancing Better Business Performance in Warehouse Management. IEEE Access, 12, pp.142419–142426. <https://doi.org/10.1109/ACCESS.2024.3470221>*

## **2. Potential robot solution**

One possible solution to the above problem is developing a robot which can operate in the warehouse to help workers check inventory and deliver items. Since it uses sensors and already has a built-in map of the area, it can navigate itself through shelves, storage areas, and delivery points, all without requiring constant monitoring by human operators. The robot can even have a camera on board for recognizing specific shelf tags, product locations, or delivery targets. By taking care of tedious processes like item transportation and locating inventory, the robot would help lighten the load on workers, improve efficiency, and prevent mistakes from occurring while working in warehouses. Since the robot operates in a fixed environment indoors, it is a suitable solution to employ at any warehouse.

## **3. How you will solve the problem (mapping, navigation, image processing, machine learning, etc)?**

To solve the problem, we will develop the robot using mapping, localization, navigation, image processing, and simple machine learning or rule-based decision-making. First, the robot will use SLAM to create a map of the warehouse environment, allowing it to understand walls, shelves, pathways, and obstacle zones. After the map is created, localization methods such as AMCL or sensor fusion can be used so the robot knows its current position inside the map. For navigation, the robot will use a global planner to calculate the main route and a local planner such as DWA or TEB to avoid nearby obstacles in real time. Image processing can be used through OpenCV to detect QR codes, ArUco markers, or object labels at pickup and drop-off stations. This allows the robot to identify the correct delivery location instead of only moving blindly to a coordinate. If machine learning is added, it can be used for object detection or improving obstacle recognition, especially in dynamic environments where people or objects move unpredictably. This approach is suitable because ROS-based navigation commonly combines SLAM, localization, path planning, obstacle avoidance, and sensor fusion as core modules for autonomous mobile robots.

## **4. What function will be needed using the ROS platform?**

The robot will primarily utilize the Navigation Stack (Nav2) to manage autonomous movement, integrating nodes for global and local costmaps to ensure safe path planning. To achieve accurate environment mapping, the SLAM Toolbox or Gmapping package will be implemented to generate a high-resolution 2D occupancy grid by processing LaserScan data from a LiDAR sensor. We will employ the AMCL (Adaptive Monte Carlo Localization) package to handle real-time robot localization, allowing the system to track its position relative to the warehouse map via a transform tree (\$tf2\$). For the delivery and inventory tasks, we will use image\_transport and cv\_bridge to interface ROS with OpenCV for marker detection, alongside ActionLib or ROS 2 Actions to manage long-running goals like moving to a specific shelf or performing a pickup sequence. Furthermore, the robot's hardware communication will be handled through ros\_control, providing a standardized interface for velocity commands (\$cmd\_vel\$) to the motor drivers and feedback from encoders.

| Function      | ROS Package/Component           | Purpose                                                  |
|---------------|---------------------------------|----------------------------------------------------------|
| Mapping       | slam_toolbox or gmapping        | Creating the warehouse floor plan.                       |
| Localization  | amcl                            | Telling the robot "where am I" on the map.               |
| Path Planning | nav2_planner or nav2_controller | Calculating the route and avoiding moving people.        |
| Vision        | cv_bridge                       | Converting ROS images into a format OpenCV can read.     |
| Coordination  | ros2_lifecycle_manager          | Managing the state of the robot (Idle, Moving, Picking). |